ROB-06

TEACHER GUIDE

Robotics · Motors & Motion Control

Servos and Steppers: Moving Precisely

A DC motor spins. A servo points. A stepper counts. Which one draws a perfect circle?

1 At a glance

LENGTH	TYPE	GROUPING	CONCEPTS
60 minutes	Hardware + software	Pairs	servo PWM · steps per revolution · open-loop

2 Overview

Students meet the two motors that move precisely. A servo points to an angle and holds it, because a potentiometer inside checks its position (closed loop). A stepper moves in fixed steps it counts but never checks (open loop). They sweep a servo, turn a 28BYJ-48 stepper exactly one revolution, build a servo gauge, then find each motor's limits and choose the right motor for a job.

> **The one idea that must land**
>
> A DC motor controls speed, a servo controls angle, a stepper counts steps. The servo checks itself; the stepper only counts, so it never knows if it skipped.

3 Learning outcomes

By the end, students can:

1. Compare **DC, servo and stepper** motors and say what each is for.
2. Explain how a **pulse width** (1 to 2 ms) sets a servo angle.
3. Drive a servo to an angle and a stepper **an exact number of steps**.
4. Pick the right motor for a job and **justify it**.

4 Three depths

Pick one for the class. Everyone does the same lesson; only the build target changes.

Starter

Servo pointer that shows the weather

Do the draw-a-square challenge and the Be the stepper game, then build the servo pointer (step 1) and make it show a 'weather' dial: sunny, cloudy, rain at three angles.

Core

Servo sweep and stepper turns exactly one revolution

Full lesson: servo sweep (step 1) and the stepper turning exactly one revolution and back a quarter (step 2), then the servo gauge if time allows.

Advanced

Micro-stepping maths and acceleration

Work out degrees per step and prove it with a protractor, add acceleration to the stepper, and try half stepping (4096 steps per turn).

5 Before the lesson

Kit, per group of 2

- SG90 micro servo with horns (orange signal, red 5V, brown GND)
- 28BYJ-48 stepper motor + ULN2003 driver board
- Arduino Uno + USB cable
- A separate 5 V supply for the stepper (USB power bank with a cut USB lead, or a 4×AA pack)
- Half-size breadboard and male-female jumper wires
- A 10 kΩ potentiometer (for the gauge in step 3), card, tape

Simulators

- **Circuit Lab**: both the servo and the stepper with ULN2003 run on screen, including step counting.
- **Wokwi** (online): ready servo and stepper examples.

Prep checklist

- [ ] Check the Servo and Stepper libraries are present (both come with the Arduino IDE).
- [ ] Test every SG90 with the sweep sketch: a stripped servo buzzes and jerks. Keep spares.
- [ ] Prepare the stepper's own 5 V supply for each pair. The stepper must not run from the Arduino's 5 V pin.
- [ ] Cut card arrows and print or draw 0 to 100 dials for the gauge.
- [ ] Clear a space for the Be the stepper game: five students standing in a square.

6 Timing plan

Min	Phase	What happens
4	Let's think	Draw a square with eyes shut, then open (slide 2).
2	Aims	The four aims and keywords (slide 3).
14	Teach	Compare the three motors, servo pulses, stepper steps, real examples (slides 4 to 7).
5	Be the stepper	Coils and rotor game, then backwards and half stepping (slide 9).
5	Kit and wiring	Power warning; servo and stepper wiring (slides 10 to 12).
8	A servo that points	Sweep, feel it push back, point to a typed number (slide 13).
10	A stepper that counts	2048 steps, tape mark comes home, then 1024 (slide 14).
5	Gauge or limits	Servo gauge with map(), or two rows of the limits table (slides 15, 16).
7	Check and exit ticket	Self-rating and the cup-to-shelf question (slides 18, 19).

Total: 60 minutes

7 Step by step

Let's think

4 min · slides 1-2

Say: Close your eyes and draw a 10 cm square. Now open them. Is it square? Now do it again with your eyes open.

Say: A DC motor draws with its eyes shut. A stepper counts its moves. A servo has a tiny eye inside that checks.

→ Students guess which motor is in a printer, a toy car, a camera gimbal and a robot arm elbow, and write it down.

Aims

2 min · slides 3

Say: Today you make one motor point and another count, and you will know which to pick for any machine.

Teach

14 min · slides 4-7

Say: A servo reads the width of a pulse sent every 20 ms. 1 ms means 0°, 1.5 ms means 90°, 2 ms means 180°.

Say: Inside the servo a potentiometer reports the real angle; the controller compares it with the pulse. Push the arm and it pushes back. That is closed loop.

Say: A stepper has four coils switched in turn; each switch is one step. The 28BYJ-48 has 32 steps per rotor turn and a 64:1 gearbox, so 2048 steps per output turn.

Say: If a stepper is overloaded it skips a step and nothing tells it. That is open loop. 3D printers use a home switch for this reason.

→ Slide 4: read the table row by row; ask which row matters for a robot arm.

→ Slide 6: work out 360 ÷ 2048 = 0.176° per step on the board.

Look for: students saying the servo 'counts' degrees. It measures them with the potentiometer; the stepper is the one that counts.

Be the stepper

5 min · slides 9

Say: You four are coils. You in the middle are the rotor: point at whoever I light up.

- Call 1, 2, 3, 4 faster and faster until the rotor cannot keep up: a skipped step.
- Call 4, 3, 2, 1: it reverses.
- Light two neighbours at once: the rotor points between them (half stepping).

Kit and wiring

5 min · slides 10-12

Say: One SG90 on the Arduino's 5 V is fine. The stepper always takes its own 5 V. Share the ground, never the 5 V.

- Servo: orange to pin 9, red to 5V, brown to GND.
- ULN2003: IN1→8, IN2→9, IN3→10, IN4→11; its + and – to the separate 5 V supply; its – also to Arduino GND.

Look for: the stepper powered from the Arduino 5V pin. It may work for a moment, then the board resets.

A servo that points

8 min · slides 13

Say: `arm.write(90)` takes degrees. The library turns them into the right pulse for you.

- Upload the sweep; watch 0 to 180 and back.
- Hold the arm gently: feel it push back.
- Extension: read a number from the Serial Monitor with `Serial.parseInt()` and write it to the servo.

Look for: anyone twisting a powered servo by hand. That strips the plastic gears; cut the power first.

A stepper that counts

10 min · slides 14

Say: The pins are wired 8, 9, 10, 11, but the Stepper library wants them in the order 8, 10, 9, 11 for this board. That is not a typo.

- Put a tape flag on the shaft.
- Run 2048 steps: the flag comes home. Then 1024: half a turn.
- Keep `setSpeed` at 10 to 15 rev/min.

Look for: a motor that hums and vibrates without turning. The pin order in the sketch is wrong, or the supply is too weak.

Gauge or limits

5 min · slides 15-16

Say: `map()` stretches a sensor's range onto the range your output needs, in one line.

- Faster pairs build the gauge with a potentiometer on A0.
- Others try two rows of the limits table and record what happened, including the speed where the stepper first skips.

Check and exit ticket

7 min · slides 18-19

Say: Give a reason, not just a name. 'Servo, because it holds an angle under load' scores; 'servo' alone does not.

- Self-rate against the proofs.
- Exit ticket: lift a cup to a shelf and hold it there. Which motor, and what happens if you choose wrong?
- Return to the guesses from the start.

8 Common misconceptions

Students may think	How to help
A servo can spin round and round.	A standard SG90 turns only 0 to 180°. (Continuous-rotation servos exist but lose position control.)
The voltage sets the servo angle.	The pulse width sets it: 1 ms to 2 ms, repeated every 20 ms. The supply just powers the motor.
A stepper knows where it is.	It only knows how many steps it was told to take. A skipped step is invisible to it.
Faster stepping is always better.	A stepper loses torque as it speeds up; too fast and it skips or stalls.
The Stepper pin order 8, 10, 9, 11 is a mistake.	It matches how the library energises the coils for this driver. Wired 8, 9, 10, 11, the sketch must list 8, 10, 9, 11.

9 Troubleshooting

Problem	Fix
Servo jitters or buzzes	It is asked for an angle it cannot reach, something is jammed, or the 5 V is sagging. Try 10 to 170° and check nothing blocks the arm.
Arduino resets when the servo moves	Too much current from the 5V pin. Use one servo only, or give servos their own supply with grounds joined.
Stepper vibrates but does not turn	Pin order: wired IN1-IN4 to 8, 9, 10, 11, the sketch must say `Stepper(STEPS, 8, 10, 9, 11)`.
Stepper turns but loses position	Too fast or overloaded. Lower setSpeed, reduce the load.
Stepper does nothing; ULN2003 LEDs dark	No supply to the driver board, or its ground is not joined to the Arduino.
Tape mark does not return home	Steps were skipped. Slow down, then re-run 2048 steps.

10 Assessment rubric

	Emerging	Secure	Extending
Comparing motors	Names the three motors.	Says what each controls and gives a use for each.	Explains open vs closed loop and why it matters for a given machine.
Servo	Runs the sweep.	Explains pulse width and points the servo to a chosen angle.	Builds the gauge with map(), or drives the servo with raw pulses.
Stepper	Gets the stepper turning with help.	Turns exactly one revolution and back a quarter; explains 2048.	Finds the skip speed, adds acceleration or half stepping, and explains the results.

11 Answers

Slide 2: Printer, toy car, camera gimbal, robot arm elbow?

> Printer: stepper. Toy car: DC motor. Gimbal: servo (or brushless with feedback). Arm elbow: servo for small arms; it must hold an angle under load.

Slide 6: Degrees per step on the 28BYJ-48?

> 360 ÷ 2048 = 0.176°. Half stepping gives 4096 steps and 0.088°.

What does a 1.5 ms pulse mean to a servo?

> Go to the middle, 90°.

Slide 16: Why does the stepper skip at 30 rev/min?

> Its torque falls as speed rises; at high speed it cannot pull the rotor round before the next coil switches.

Exit ticket: lift a cup to a shelf and hold it.

> A servo: it holds an angle and pushes back against the load. A DC motor would let the cup drop when power stops changing; a stepper could hold, but would silently lose position if overloaded.

12 Safety

- ! Low voltage only: 5 V from USB and a separate 5 V supply for the stepper.
- ! Never turn a powered servo by hand: the gears strip.
- ! One servo on the Arduino's 5 V at most; motors with their own supply always share the ground.
- ! The stepper and ULN2003 get warm in use. That is normal; hot enough to hurt is not. Unplug and check the supply.